Nested Bandits

International Conference on Machine Learning

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July, 2022

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The Blue Bus / Red Bus Paradox (1/2)

Choosing a mean of transportation

Alternatives:

- a **car**, which takes on average 15 mins ($v_{\text{car}} = -15$)
- a **bus**, which takes on average 20 mins ($v_{\text{bus}} = -20$)

Logit choice [1, 2]

- $\mathbb{P}(\text{car}) = \frac{\exp(v_{\text{car}})}{\exp(v_{\text{car}})+\exp(v_{\text{bus}})} \approx 0.62$ most probable choice
- $\mathbb{P}(\text{bus}) = \frac{\exp(v_{\text{bus}})}{\exp(v_{\text{car}})+\exp(v_{\text{bus}})} \approx 0.38$

The Blue Bus / Red Bus Paradox (2/2)

Choosing a mean of transportation

Alternatives:

- a **car**, which takes on average 15 mins ($v_{\text{car}} = -15$)
- a **blue bus**, which takes on average 20 mins ($v_{\text{bus}} = -20$)
- a **red bus**, identical to the blue bus (except its color)

Logit choice [1, 2]

- $\mathbb{P}(\text{car}) = \frac{\exp(v_{\text{car}})}{\exp(v_{\text{car}}) + 2\exp(v_{\text{bus}})} = 0.45$ no longer most probable!
- $\mathbb{P}(\text{blue bus}) = \mathbb{P}(\text{red bus}) = \frac{\exp(v_{\text{bus}})}{\exp(v_{\text{car}}) + 2\exp(v_{\text{bus}})} = 0.27$

Problem

Logit choice no longer reasonable: an irrelevant alternative switches choice odds!

Adversarial Bandits (1/2)

Notations and incurred regret of EXP3

- $(v_{a,t})_{a \in \mathcal{A}}$ *payoff vector* of stage $t = 1, 2, \ldots T$
- $P_t(a)$ probability of choosing arm a at stage t (n arms)
- $r_t = v_{a_t,t}$ *reward* received at stage t from arm $a_t \sim P_t$

$$\text{Reg}(T) \leq \sqrt{2n \log(n) T}$$

Blue Bus / Red Bus situation

Two alternatives $a_1, a_2 \in \mathcal{A}$ generate consistently same reward:

can we avoid considering both alternatives in a bandit algorithm?

More general: $\mathcal{A}$ has an inherent structure?

If n very big but some alternatives have very similar rewards:

can we exploit this side information to design a more efficient algorithm?

Adversarial Bandits (2/2)

Notations and incurred regret of EXP3

- $(v_{a,t})_{a\in\mathcal{A}}$ *payoff vector* of stage $t = 1, 2, \dots T$
- $P_t(a)$ probability of choosing arm a at stage t (n arms)
- $r_t = v_{a_t,t}$ *reward* received at stage t from arm $a_t \sim P_t$

$$\mathrm{Reg}(T) \leq \sqrt{2n \log(n) T}$$

Nested Exponential Weights algorithm

If we exploit side-information on the structure of $\mathcal{A}$ and regularity of $(v_a)_{a\in\mathcal{A}}$, we propose to use the **Nested Exponential Weights** (NEW) algorithm to obtain

$$\mathrm{Reg}(T) \leq \sqrt{2n_{\mathrm{eff}} \log(n) T}$$

where n_{eff} is typically much smaller than n and we always have $n_{\mathrm{eff}} \leq n$.

General Similarity Model

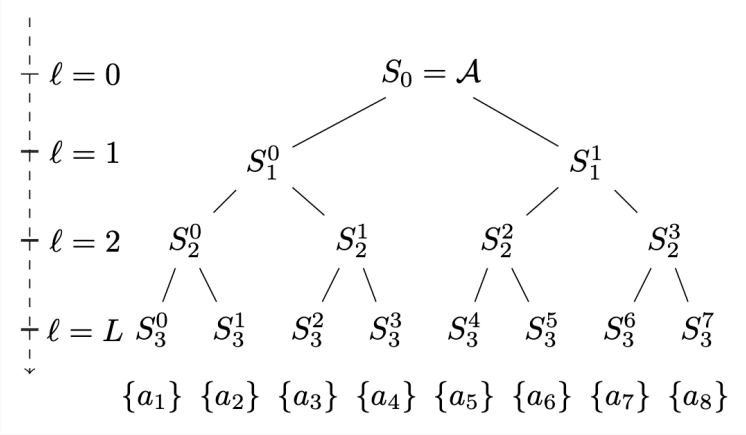


Figure 1: Nested structure: ($L = 3$)

- $\mathcal{A} := \{a_i : i = 1, \dots, n\}$ set of *alternatives*
- $\{\mathcal{A}\} =: \mathcal{S}_0 \succcurlyeq \cdots \succcurlyeq \mathcal{S}_L := \{\{a\} : a \in \mathcal{A}\}$ tower of *partitions*

Reward & Feedback

For all $a \in \mathcal{A}$ and $a \equiv S_L \lhd S_{L-1} \lhd \cdots \lhd S_0 \equiv \mathcal{A}$ its *lineage*,

$$v_a = \sum_{\ell=1}^{L} r_{S_\ell}$$

Semi-bandit feedback: at each round, the learner *observes each* r_{S_ℓ}

$$r_{S_\ell} \in [0, R_\ell] \quad \text{for all } S_\ell \in \mathcal{S}_\ell,\ \ell = 1, \dots, L,$$

where $R_\ell \geq 0$ represents the *reward variability* for $\mathcal{S}_\ell$

Nested Exponential Weights

Algorithm

For each stage $t = 1, 2, \ldots,$ given $y_t \in \mathbb{R}^{\mathcal{A}}$ (current score), η_t (learning rate) and μ_ℓ (uncertainty level parameter), the learner:

1. computes choice probability P_t from **Nested Logit Choice** (NLC) $P_{S_\ell|S_{\ell-1}}(y)$ and y_t using **upward pass** on level scores y_{S_ℓ}

$$P_{S_\ell|S_{\ell-1}}(y) = \frac{\exp(y_{S_\ell}/\mu_\ell)}{\exp(y_{S_{\ell-1}}/\mu_\ell)} \qquad \text{(NLC)}$$

2. selects action $a_t \in \mathcal{A}$ following **downward pass** in (NLC)

$$a_t \sim P_t(\eta_t y_t)$$

3. uses level rewards $r_{S,t}$ for each class $S \ni a_t$ and constructs a **Nested Importance Weighted Estimator** (NIWE) $\hat{v}_t$ of the payoff vector of stage t
4. updates their score: $y_{t+1} \leftarrow y_t + \hat{v}_t$ and the process repeats

Regret guarantees for NEW

Theorem

Defining $\sqrt{n_{\text{eff}}} = \sum_{\ell=1}^{L} \sqrt{n_\ell} R_\ell$, *if NEW is run with* $\eta_t = \sqrt{\log n/(2t)}$, *we have*

$$\mathbb{E}[\text{Reg}_p(T)] \leq 2\sqrt{2n_{\text{eff}} \log n \cdot T}.$$

Comparison to EXP3

Regret guarantees of NEW and EXP3 differ by a factor of

$$\alpha = \sqrt{n/n_{\text{eff}}},$$

Suppose red bus / blue bus problem with

- $n_1 = 2$ classes and $n_2 = 100$ alternatives per class
- negligible intra-class reward differential ($R_2 \approx 0$)

regret guarantees improves by a factor of $\alpha \approx 10$

Benefits in the red bus / blue bus problem

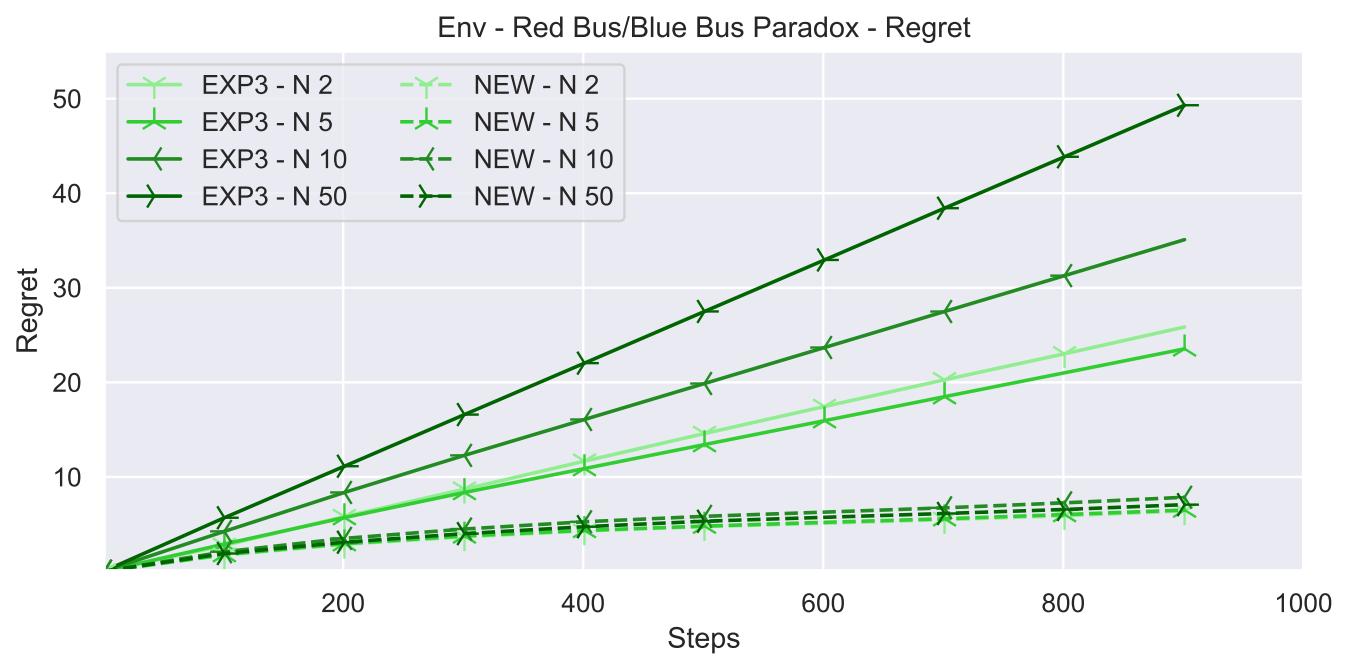


Figure 2: Regret of EXP3 and NEW in the red bus / blue bus problem with different numbers of buses N.

Interpretation

NEW systematically achieves better regret than EXP3 and is far less sensible to N

Conclusion

The **Nested Exponential Weights** (NEW) algorithm combines:

- the **Nested Logit Choice** (NLC) rule
- the **Nested Importance Weighted Estimator** (NIWE)

resulting in an improved adversarial bandit algorithm exploiting side-information on the **structure** of $\mathcal{A}$ and **regularity of** $(v_a)_{a \in \mathcal{A}}$

Thank you!

References

[1] R. D. Luce. *Individual Choice Behavior: A Theoretical Analysis*. Wiley, New York, 1959.

[2] D. L. McFadden. Conditional logit analysis of qualitative choice behavior. In P. Zarembka, editor, *Frontiers in Econometrics*, pages 105–142. Academic Press, New York, NY, 1974.